

A Three-Window Decomposition of Cosmological Redshift in Energy-Flow Cosmology

A synthesis note: T_{CMB} , CDDR , and $S(z)$ consistency under one regime parameter

Morten Magnusson

ORCID: [0009-0002-4860-5095](https://orcid.org/0009-0002-4860-5095)

Symbiose Research, Norway

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Abstract. Within Energy-Flow Cosmology (EFC), observed cosmological redshift admits a decomposition into three physically distinct contributions sourced by a single effective entropy field S_{eff} derived from the EFC primitives $\{S_{\text{flow}}, S_{\text{latent}}\}$ via the screening parameter ξ . We formalise the decomposition as a single line integral, identify three observational windows (thermal, geometric, structural) that probe different functional dependencies of $\mathcal{F}(\alpha, S_{\text{eff}}, \nabla S_{\text{eff}})$, and show that all three reduce to ΛCDM in the limit $\nabla S_{\text{eff}} \rightarrow 0$. The framework links the three to one regime-coupling scale a_G via simple rational projections established elsewhere in the EFC literature ($a_c = a_G/4$, $a_{H_0} = a_G/2$). This is a synthesis of existing EFC results, not a new derivation; the integral form is offered as a unifying notation for the three approved validation tests in the EFC ledger. Two open tensions are flagged explicitly: the a_0 -discrepancy ($\sim 5\times$ between cubic-graph prediction and SPARC calibration) and the unresolved cosmological-vs-galactic regime split of a_G .

1 Definition

Let γ denote a null geodesic connecting an emission event at affine parameter λ_e and an observation event at λ_o in a spacetime with EFC primitive fields $\{\alpha, S_{\text{flow}}, S_{\text{latent}}, \xi, K, T\}$. The cosmological redshift is then expressed as

$$\ln(1 + z_{\text{obs}}) = \int_{\gamma} \mathcal{F}(\alpha, S_{\text{eff}}, k^{\mu} \nabla_{\mu} S_{\text{eff}}) d\lambda, \quad (1)$$

where $k^{\mu} = dx^{\mu}/d\lambda$ is the photon four-momentum along γ , and the integrand $\mathcal{F}$ is a dimensionless scalar functional of α , S_{eff} , and the directional derivative $k^{\mu} \nabla_{\mu} S_{\text{eff}}$ taken along the photon path. The remaining EFC primitives $\{K, T\}$ enter implicitly through the α -coupling and need not appear explicitly in $\mathcal{F}$ at the level of the redshift decomposition. Equation (1) replaces the metric-only relation $1 + z = a_o/a_e$ of ΛCDM with a path-dependent form that admits structure-dependent contributions along the line of sight, while remaining compatible with ΛCDM as a strict limit (Section 5).

Physically, Equation (1) expresses the accumulated change in photon energy along γ as a functional of the entropy field configuration encountered along the path. We emphasise that $\mathcal{F}$ is *not* an arbitrary function: its functional form is constrained by the single scale a_G (Section 3) and must simultaneously satisfy all three observational windows (Section 4) without introducing additional free parameters beyond those already present in EFC.

2 Effective field

EFC carries two distinct entropy fields: S_{flow} (driving cosmic expansion at L0–L1) and S_{latent} (driving perturbation-sector dynamics at L2–L3). The screening parameter ξ couples them. We define the effective entropy field for redshift propagation as the projection

$$S_{\text{eff}} = \mathcal{P}_{\xi}(S_{\text{flow}}, S_{\text{latent}}), \quad (2)$$

where $\mathcal{P}_{\xi}$ is the regime-dependent projection operator that recovers S_{flow} in the unscreened cosmological regime and admits S_{latent} -corrections inside collapsed structures via the ξ -screening scale. A canonical first-order form consistent with existing EFC perturbation-sector results is the additive projection

$$S_{\text{eff}} \approx S_{\text{flow}} + \xi S_{\text{latent}}, \quad (3)$$

valid where ξ varies slowly along γ . Equation (3) is *not* a fundamental definition but a first-order approximation valid in regimes where ξ is approximately constant along the photon path; it preserves the two-field structure required by the sealed perturbation-sector predictions while reducing the redshift integrand to a one-field problem along γ .

3 Single regime scale

The functional dependence of $\mathcal{F}$ on S_{eff} is anchored to a single underlying structural scale $a_G \approx 0.1$ (the gravitational coupling coefficient of EFC, possibly regime-dependent), with regime-specific projections established elsewhere in the framework. Observable amplitudes in the propagation sector may, however, involve additional effective parameters (e.g. the ϵ , β of the $c_{\text{eff}}(a)$ parameterisation, with β itself anchored to BAO+RSD data) that are not reducible to a_G alone. The structural projections below are exact relations on the underlying scale; the mapping to observable amplitudes is sector-specific and remains open in places.

Regime / phenomenon	Projection of a_G	Source
Hubble structure (Friedmann)	$a_{H_0} = \frac{1}{2}a_G$	$H^2 \propto G_{\text{eff}}$ structure
Photon propagation (Core Lock)	$a_c = \frac{1}{4}a_G$	$\ln(c/c_0) = a_c \Phi(S)$
Galactic acceleration (MOND-like)	$g/a_0 \approx 5$	SPARC anchoring (open)
Perturbation amplitude	$C_{\text{eff}} = k^2/a_G$	EFC-D action level

Table 1: Regime-specific projections of the irreducible scale a_G . The MOND-line is flagged as open (Section 7).

This is a structural claim, not a numerical assertion that one universal value of a_G fits all regimes; the cosmological-versus-galactic consistency of a_G is treated as an open question in Section 7.

4 Observational windows

Equation (1) admits three observational probes, each sensitive to a different functional dependence of $\mathcal{F}$ on S_{eff} :

The ledger entries in question are formal **EFCValidation** nodes with status **planned** as of the date of this note. Crucially, the windows are *not* independent degrees of freedom: they are projections of the same integrand onto different observables. A single a_G value (modulo the open question of regime-dependence) must fit all three, or the framework is internally inconsistent.

Window	What it constrains in $\mathcal{F}$	Test ID in EFC ledger
Thermal $T_{\text{CMB}}(z)$	$\partial_S \mathcal{F}$ via $T(z)$ -scaling	<code>conv_t_cmb_z_deviation...</code>
Geometric CDDR $\eta(z)$	transport sector via $\Phi(S)$	<code>conv_c_eff_a_via_cddr...</code>
Structural JWST $\leftrightarrow$ CMB	integral of $S(z)$ via consistency	<code>conv_s_z_single_parameter...</code>

Table 2: The three approved data-driven validation tests in the EFC ledger map onto three distinct functional probes of $\mathcal{F}$.

4.1 Order-of-magnitude expectation for the CDDR window

At the closure value $a_c = a_G/4 \approx 0.024$, Core Lock gives $c_{\text{eff}}/c_0 \approx \exp(a_c \Phi_{\text{max}}) \approx \exp(0.024)$ at maximum traversed entropy gradient, which translates to a redshift-integrated CDDR deviation

$$|\eta(z) - 1| \sim 10^{-3} - 10^{-2} \quad (\text{preferred range}), \quad (4)$$

with the upper envelope reaching $|\eta - 1| \sim 5 \times 10^{-2}$ in extreme saturation regimes of $\Phi(S)$. The preferred range is consistent with current observational bounds (Wang et al. 2025; Liu et al. 2026) at low redshift, and overlaps the marginal positive-evidence regime ($\gtrsim 2\sigma$) reported at $z \sim 2.3$ in DESI DR2 + Pantheon analyses. We emphasise that M_B -calibration is the dominant systematic confounder in CDDR tests and any falsification claim derived from Equation (1) must be conditioned on M_B -free analysis.

5 Λ CDM limit

In the limit $\nabla S_{\text{eff}} \rightarrow 0$ along γ , the integrand $\mathcal{F}$ reduces to its purely geometric component:

$$\mathcal{F}(\alpha, S_{\text{eff}}, \nabla S_{\text{eff}} \rightarrow 0) \longrightarrow \frac{d \ln a}{d\lambda}, \quad (5)$$

recovering the standard relation $\ln(1+z) = \int d \ln a$. All three observational windows yield null deviations in this limit, and the framework collapses to Λ CDM exactly. This is a feature, not a weakness: it means the framework is empirically distinguishable from Λ CDM only through the same three windows it predicts, and a triple-null result constitutes a clean falsification.

6 Perturbation-sector reduction

In the linear-response regime $\delta S_{\text{latent}} \ll \langle S_{\text{flow}} \rangle$, Equation (1) reduces to the existing EFC perturbation-sector predictions without introducing additional free parameters:

- E_G statistic: $E_G^{\text{EFC}}/E_G^{\text{GR}} = 1.086$ (sealed prediction, KILL-tagged)
- Bullet cluster gravitational slip: $\eta = \Phi/\Psi = 0.9993$ (sealed prediction, KILL-tagged)
- μ - Σ quadrant: $\mu < 1$ AND $\Sigma > 1$ (sealed prediction, KILL-tagged)
- ϕ - ψ -slip transition redshift: $z = 0.44$ (sealed prediction, KILL-tagged)
- σ_8 growth: $\sigma_8 = 0.43$ (sealed prediction, DAMAGE-tagged)

These are properties of the perturbation sector independent of the redshift-decomposition presented here. Equation (1) does not introduce additional independent degrees of freedom that would weaken or alter these predictions; it organises the L0–L1 background-and-propagation sector into a form whose null limit yields Λ CDM and whose first-order corrections are constrained by the three approved tests.

7 Open tensions

Two unresolved issues are flagged explicitly:

The a_0 -discrepancy

The cubic-graph derivation of EFC predicts an effective MOND scale $a_{0,\text{eff}} \approx 5.4 \cdot a_0^{\text{SPARC}}$, where $a_0^{\text{SPARC}} \approx 1.2 \times 10^{-10} \text{ m/s}^2$ is the empirical calibration. This $\sim 5\times$ gap is at kill-test level and is *not* addressed by the present framework. Until it is resolved, any unification claim in Section 3 must be qualified: a common scale a_G is structurally required, but its numerical consistency across regimes is empirically open.

Cosmological vs galactic regime split of a_G

The EFC graph carries two distinct nodes, “ a_G Cosmological” (cubic-graph derivation) and “ a_G Galactic” (SPARC anchor). These are presumed numerically equal at ~ 0.1 but their independent calibration is incomplete. A laboratory anchor (e.g. via Casimir-scale or torsion-balance experiments sensitive to ξ -modulated propagation) would close the consistency gap. Without it, the framework’s claim of a universal a_G remains a structural hypothesis rather than an established result.

8 Status

This note is a **synthesis**, not a derivation. Equation (1) is offered as a unifying notation that organises three approved EFC validation tests under one functional form. The integral expression is not derived from a Lagrangian here, nor is the projection operator $\mathcal{P}_\xi$ specified beyond its first-order linear approximation. The note’s contribution is structural: identifying that the three approved tests probe one a_G -parameterised integrand, with falsifiability preserved through the Λ CDM limit and consistency required across all three observational windows.

The synthesis is anchored in the EFC Neo4j knowledge graph (v2026.04) and the Frameworks-Atlas GNN embeddings, which return similarity scores > 0.84 between the a_G , a_0 , and Core Lock concept nodes — consistent with the regime-projection structure used in Section 3.

The precise form of $\Phi(S)$ or $T(a)$ and the commitment to an optical-medium interpretation of c_{eff} (Case ii in the underlying $c_{\text{eff}}(a)$ paper, DOI 10.6084/m9.figshare.31305421) remain open theoretical questions in the EFC literature and are not resolved in this synthesis. The CDDR window in Section 4.1 is therefore conditional on Case (ii); the metric-VSL alternative (Case i) gives $\eta(z) = 1$ identically and removes one of the three observational windows. Resolving this commitment is a prerequisite for any operational form of $\mathcal{F}$.

The framework is empirically actionable because the three tests can be run on existing data without new observational time:

- Pantheon+ (D_L) $\times$ DESI DR2 BAO (D_A) $\times$ galaxy-cluster X-ray+SZ ladder, for CDDR;
- SZ-cluster compilations (Hurier+2014, Luzzi+2015, Saro+2014) for $T_{\text{CMB}}(z)$;
- JWST high- z luminosity function (Labbé+2023, Harikane+2024, CEERS-2024) joint with the SZ datasets for $S(z)$ consistency.

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EFC validation ledger entries referenced:

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- conv_s_z_single_parameter_joint_fit_cmb_anomaly_jwst_early_galaxy
- conv_c_eff_a_via_cddr_violation_d_l_vs_d_a_1_z_consistency_test